

regional monitoring. Although all eyes are now trained on LEO satellites, due to the promise of affordable, scalable, low-latency, high-speed connectivity across diverse geographies, a country like India would typically need a multi-orbit approach, together with existing terrestrial communication networks to bridge the digital divide reliably and cost-effectively.

Multi-orbit strategies of global players

Satcom players are opting for multi-orbit strategies to diversify their product and service offerings. We see industry leaders consolidating GEO, MEO, and LEO assets through expansion and acquisitions.

Viasat and Inmarsat, which it acquired in 2023, together have a formidable GEO satellite setup, which they use to provide residential, government, enterprise, in-flight, and maritime connectivity solutions. To counter the competition from Starlink, Viasat recently signed an agreement with Telesat to integrate its upcoming Telesat Light-speed LEO Ka-band capacity into its multi-orbit network. According to Viasat, integrating a flexible and resilient LEO constellation with its own ultra-high throughput GEO satellites will make it possible to come up with superior yet cost-effective differentiated broadband connectivity solutions.

Luxembourg-based SES is a pioneer of the multi-orbit strategy. It has been running the O3b mPOWER MEO constellation alongside its GEO fleet for over a decade now. While GEO enables reliable video distribution and data services, the MEO constellation enables high-throughput, low-latency connectivity comparable to terrestrial fibre in performance. Apart from a minor collaboration with Starlink for a connectivity solution for cruise ships, SES has

Local-global synergy spells opportunities aplenty

- Traditional telecom players can partner with satcom players to boost the reach of their terrestrial networks
- Global satcom players are seeking to work with local internet service providers (ISPs) for sales and service of their satellite connectivity products for consumers
- LEO-focused companies like Starlink, Amazon Kuiper, and Globalstar are seeking partnerships with established VSAT companies in India to offer enhanced, multi-orbit connectivity and mobility solutions to enterprise and government segments
- As satcom entrants start setting up their ground infrastructure, there will be a boost in demand for components like transmitters/transponders, receivers, antennae, and transceivers
- Good time to foray into standardised satellite manufacturing (as LEO satellites must be replaced every 5-10 years)

predominantly focused on a GEO plus MEO strategy, even emphasising on their website that it works best for their clients.

Recently, however, it has exhibited a greater commitment towards including LEO in its multi-orbit strategy. SES and Lynk Global have inked a pact to share the former's GEO/MEO satellites and ground infrastructure and the latter's direct-to-device (D2D) capabilities and growing constellation of LEO satellites.

According to news reports, SES has also reinstated its filing for its Cleosat LEO scheme with the International Telecommunications Union (ITU), which may be a consequence of Europe's push to promote regional players and reduce reliance on global satellite players, such as SpaceX and Amazon, amid concerns about digital sovereignty.

SES is also in the final phases of acquiring Intelsat, a pioneer in satellite technology. Intelsat operates one of the world's largest and most advanced GEO satellite fleets and terrestrial connectivity infrastructure and is developing MEO and LEO infrastructure as part of its multi-orbit strategy.

GEO major Eutelsat nailed its multi-orbit strategy by acquiring OneWeb in 2023. This forges Eutelsat's fleet of GEO satellites with OneWeb's fully deployed LEO constellation, enabling a wide range of offerings in video broadcasting,

enterprise and government connectivity, cellular backhaul, and mobility.

And that brings us to SpaceX's Starlink, whose foray into the Indian market has suddenly piqued everyone's interest in direct-to-consumer satellite broadband services! Starlink is entirely focused on LEO satellites. It already owns more than half the LEO satellites in space and shows no signs of slowing down!

SpaceX's Starlink focuses on direct-to-consumer broadband, enterprise solutions, and mobility for recreational vehicles and maritime vessels. It operates in around 100 countries. Its Starshield business unit runs dedicated military and intelligence programmes, mainly for the US and its allies.

Wooing the Indian broadband consumers

Almost all global majors have entered the Indian satellite broadband market in collaboration with one or more local players. While some, like Starlink, are proposing direct-to-consumer services, others are looking at complementing existing telecommunication systems with cellular backhaul to extend their reach to remote areas; or provide backup communication systems during emergencies and natural disasters.

BSNL with Viasat. Viasat, which has a good B2B presence in India, is looking at expanding to